

## Refraction of Light

### I] Textbook Questions:

#### ★ 1 Mark Objectives: ★

1. Mark the correct answer in the following questions.

A. What is the reason for the twinkling of stars?

- i . Explosions occurring in stars from time to time
- ii. Absorption of light in the earth's atmosphere
- iii. Motion of stars

iv. Changing refractive index of the atmospheric gases

B. We can see the Sun even when it is little below the horizon because of

[2024]

- i. Reflection of light
- ii. Refraction of light
- iii. Dispersion of light
- iv. Absorption of light

C. If the refractive index of glass with respect to air is  $\frac{3}{2}$ , what is the refractive index of air with respect to glass?

[2023]

- i .  $\frac{1}{2}$
- ii. 3
- iii.  $\frac{1}{3}$
- iv.  $\frac{2}{3}$

2. Fill in the blanks and Explain the completed sentences.

a . Refractive index depends on the **velocity** of light.

**Explanation:** The refractive index of a medium is inversely proportional to the velocity of light. It means as the refractive index of a medium increases, the speed of light going through that medium decreases.

b. The change in **direction** of light rays while going from one medium to another is called refraction.

**Explanation:** The direction of rays of light changes when it enters from one transparent medium to another as the velocity of light is different in different media.

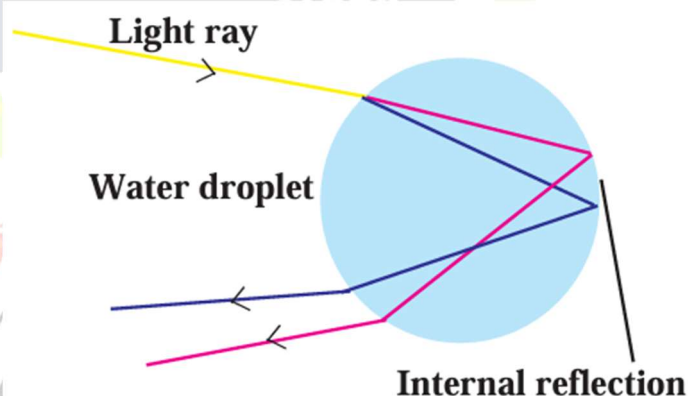
### ★ 5 Marks Questions: ★

1. Prove the following statements.

a. A rainbow is the combined effect of the refraction, dispersion, and total internal reflection of light

[2023, 2024, 2025, 2026]

- Rainbow is a beautiful natural phenomenon. It is the combined effect of a number of natural processes.
- It is the combined effect of dispersion, refraction and total internal reflection of light. It can be seen mainly after a rainfall.
- Small droplets of water act as small prisms.
- When light rays from the Sun enter these droplets, it gets refracted and dispersed.
- Then there is internal reflection as shown in the figure, and after that once again the light gets refracted while coming out of the droplet.
- All these three processes together produce the rainbow.



## II] Extra - Previous Years' Questions: (PYQs)

### ★ 1 Mark Objectives: ★

#### MCQs:

- Light Changes its direction when going from one transparent medium to another transparent medium. This process is called \_\_\_\_\_. [2025]  
(A) Reflection (B) Dispersion (C) Scattering (D) Refraction
- If a ray of light passes from a denser medium to a rater medium in a straight line, the angle of incidence must be \_\_\_\_\_. [2023]  
(A)  $0^\circ$  (B)  $30^\circ$  (C)  $60^\circ$  (D)  $90^\circ$
- \_\_\_\_\_ colour is deviated the least, in the spectrum of white light obtained with a glass prism [2022]

- (A) Red (B) Yellow (C) Violet (D) Blue

4. For a particular value of  $i$ , the value of  $r$  becomes equal to  $90^\circ$ . This value of  $i$  is called the \_\_\_\_\_ [2022]

- (A) Critical angle (B) Angle of deviation (C) Angle of refraction (D) Angle of emergence

5. The refractive index of diamond is \_\_\_\_\_ [2026]

- (A) 2.42 (B) 1.54 (C) 1.36 (D) 1.50

6. \_\_\_\_\_ has the highest refractive index.

[2023]

- (A) Air (B) Water (C) Glass (D) Diamond

7. **True or False:** Wavelength of red light is close to 700nm. (True) [2023, 2025]

8. **Correlation:** Red light: 700nm :: violet light: 400nm [2025]

9. **Match the correct pair:** [2022, 2023, 2024]

| Column I |                           | Column II |                                  |
|----------|---------------------------|-----------|----------------------------------|
| i.       | Refractive index of water | a.        | 1.31                             |
| ii.      | Air                       | b.        | 1.36                             |
|          |                           | c.        | 1.33 (refractive index of water) |
|          |                           | d.        | 1.46                             |
|          |                           | e.        | 1.0003 (air)                     |

10. Answer the following.

1. **What is the reason of twinkling of stars?** [2025]

**Ans:** Due atmospheric refraction the refractive index of the atmosphere keeps on changing and stars appear to be twinkling.

2. **What is the wavelength of violet ray in dispersion of light?** [2025]

**Ans:** The wavelength of violet ray in dispersion of light is 400nm.

3. **What is the angle of refraction when the angle of incidence is  $0^\circ$ ?** [2024]

**Ans:** The angle of refraction is  $0^\circ$ .

**2 Marks Questions:**

1. **State the laws of refraction** [2022, 2021]

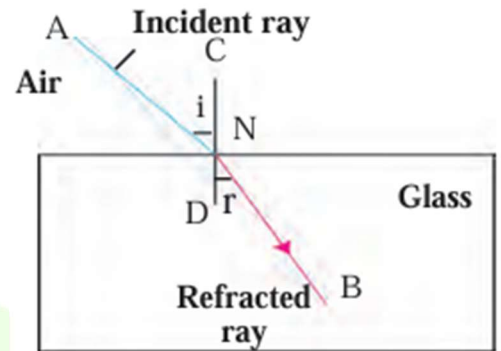
**Ans:****Laws of refraction**

1. Incident ray and refracted ray at the point of incidence N are on the opposite sides of the normal to the surface of the slab at that point i.e. CD, and the three, incident ray, refracted ray and the normal, are in the same plane.

2. For a given pair of media, here air and glass, the ratio of  $\sin i$  to  $\sin r$  is a constant called as the refractive index of second medium with

respect to the first medium.  $\frac{\sin i}{\sin r} = \text{constant} = n$

Here,  $i$  is the angle of incidence and  $r$  is the angle of refraction. This second law is also called Snell's law.

**2. Why does the apparent position of stars keeps changing a bit?**

[2023]

**Ans:**

1. The apparent position of the star keeps changing a bit.
2. This is because of the motion of atmospheric air and changing air density and temperature.
3. Because of this, the refractive index of air keeps changing continuously.
4. Because of this change, the position and brightness of the star keep changing continuously and the star appears to be twinkling.

**3. Give Reason:****We can see the sun before it emerges above the horizon.**

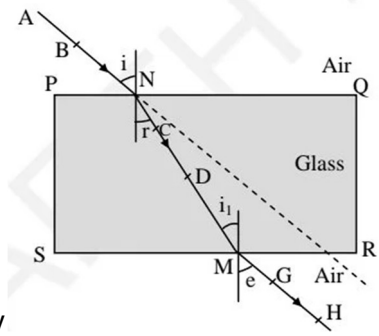
[2022]

**Ans:**

1. By Sunrise we mean the appearance of the Sun above the horizon.
2. But when the Sun is somewhat below the horizon, its light rays are able to reach us along a curved path due to their refraction through earth's atmosphere.
3. Thus, we see the Sun even before it emerges above the horizon.
4. Same thing happens at the time of Sunset and we keep seeing the Sun for a short while even after it goes below the horizon.

**3 Marks Questions:****1. Observe the figure and name the ray AB, ray CD, ray GH:**

[2023]



**Ans:** ray AB: Incident Ray, ray CD: Refracted Ray, ray GH: Emergent Ray

5. Observe the given diagram and answer the questions.

[2022, 2024]

1. Name the process shown.

**Ans:** Dispersion of light

2. Name the colour that deviates the most.

**Ans:** Violet

3. Name the colour that deviates the least.

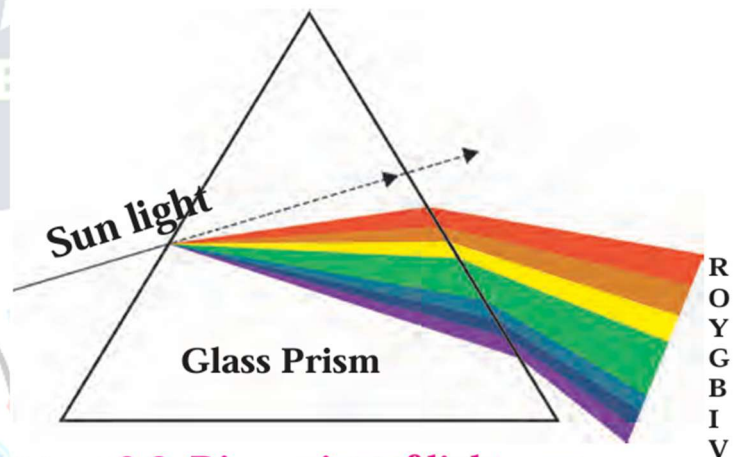
**Ans:** Red

4. Name any one phenomenon in the nature which is based on the given process.

**Ans:** Formation of Rainbow.

5. Define spectrum.

**Ans:** The band of coloured components of a light beam is called spectrum.



2. What is meant by mirage? In which condition is mirage seen? Give such one more example.

[2026]

**Ans:**

1. The light rays coming from a distant object appear to be coming from the image of the object inside the ground. This is called a mirage.

2. A mirage is usually seen:

1. On hot days, especially in deserts or on tar roads
2. When the ground is very hot, and the air just above it is warmer than the air higher up
3. This temperature difference causes light to bend (refraction), creating the illusion of water

3. Example:

1. A common example of a mirage is the appearance of a "lake" on a hot desert road, which is actually a refracted image of the sky.
2. On roads during summer, a similar effect can be observed making the road appear wet or reflective.



**3. Observer the given diagram and answer the following questions:**

[2023]

**a. What is refraction of light?**

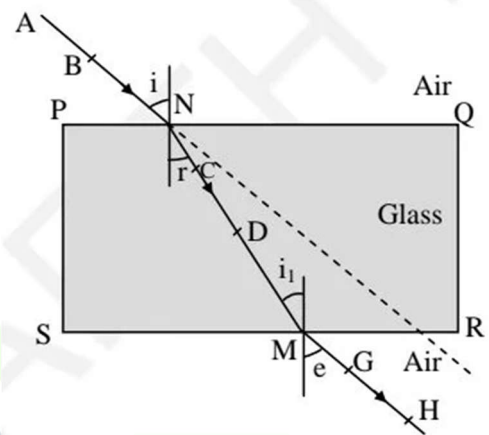
**Ans:** Light changes its direction while going from one transparent medium to other. This is called refraction of light.

**b. Name the emergent ray.**

**Ans:** ray GH is the emergent ray.

**c. Which two angles are equal?**

**Ans:**  $\angle i = \angle e$  and  $\angle r = \angle i_1$



**MASTER CLASSES**